Mid-Semester Progress Report

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NBA Game Outcome Prediction (Classification)

Based on feedback, I decided to narrow the project down to just one task instead of two. My original proposal had both a game outcome model and a player points model. I chose to focus on game outcome classification.

Progress:

- I downloaded and cleaned the Kaggle NBA dataset (the Games and TeamStatistics files). After filtering to regular season games from 2018 on, I had about 10,000 games to work with.
- I built features that only use games that happened before each matchup, so the model never sees the future. These include each team's average stats over their last 5 and 10 games, days of rest, back to back games, and their season win percentage so far.
- I reshaped the data so each game is one row, using the difference between the home team and away team for each feature.
- I set up a train/test split based on time. The model trains on older games and gets tested on the most recent season.
- For the models, I used two standard algorithms from the scikit-learn library: logistic regression and random forest.

Challenges faced:

- The biggest challenge was avoiding data leakage. I had to make sure each feature only used past games, not the current game or any future ones. Getting the averages to line up correctly by team and by date took some trial and error.
- Turning the two rows per game (one for each team) into a single row with different features was also tricky and took a few tries to get right.

Next steps: I plan to add features that adjust for how strong each team's opponents have been, tune the model settings, and test the models. (as well as making the project report)